

Lab Title: Week 12 Assignment 1

Student Name: Kabir Ibrahim

Student ID: C11/26/FCDF/17162

Instructor Name: Aminu Idris

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Version: Linux Kali 6.18.12+kali-amd64.

Part A Core Concepts

1). Define an IPv4 address and explain why the address must be unique on the same network.

An IPv4 address is a 32-bit numerical identifier that is separated by dots into 4 octets, assigned to a device, making it capable of connecting to a network using the Internet Protocol. Example 192.168.1.1.

It must be unique on the same network to ensure that data packets are correctly routed to and from the intended host without collision or delivery errors. If two devices share the same address, an IP address conflict occurs, causing network instability and preventing proper communication for both hosts.

2). State one difference between flat and hierarchical addressing. Give one example of each type.

The difference is that flat addressing lacks any internal structure or logical grouping. While hierarchical addressing divides addresses into multiple structured levels to facilitate efficient data routing.
example of a flat address is a MAC (Media Access Control) address.
example of a hierarchical address is an IP address.

3). Compare static and dynamic IP addressing. Name the protocol normally used to assign an address dynamically.

Static IP addressing involves manually assigning a permanent IP address to a network interface, while dynamic IP addressing automatically assigns a temporary address from a pool for a specific lease period.

The protocol normally use to assign IP addresses dynamically is DHCP (Dynamic Host Configuration Protocol).

Part B Classful Address Identification

IPv4 address	Class	Network and host pattern	Default subnet mask
23.14.6.9	Class A	N.H.H.H	255.0.0.0
145.80.12.200	Class B	N.N.H.H	255.255.0.0
198.51.100.27	Class C	N.N.N.H	255.255.255.0
9.200.10.1	Class A	N.H.H.H	255.0.0.0
188.44.9.72	Class B	N.N.H.H	255.255.0.0

Part C Network and Broadcast Analysis

1). 34.72.6.19

- Network address: 34.0.0.0
- Broadcast address: 34.255.255.255
- First usable host: 34.0.0.1
- Last usable host: 34.255.255.254
- Maximum usable hosts: 16,777,214 ($2^{24} - 2$)

2). 150.20.7.200

- Network address: 150.20.0.0
- Broadcast address: 150.20.255.255
- First usable host: 150.20.0.1
- Last usable host: 150.20.255.254
- Maximum usable hosts: 65,534 ($2^{16} - 2$)

3). 203.0.113.44

- Network address: 203.0.113.0
- Broadcast address: 203.0.113.255
- First usable host: 203.0.113.1
- Last usable host: 203.0.113.254
- Maximum usable hosts: 254 ($2^8 - 2$)

Part D Applied Troubleshooting

1). Give the network address and broadcast address for the workstation.

Network address = 201.110.213.0

Broadcast address = 201.110.213.255

2). Is the configured default gateway valid for this workstation? Explain your answer.

The default gateway is not valid.

The workstation's subnet consists of IP addresses from 201.110.213.1 through 201.110.213.254. The gateway address (201.110.214.1) belongs to a different network segment (201.110.214.0/24), making it unreachable without local routing.

3). Give one valid default gateway address. Explain why the network and broadcast addresses cannot be used as the gateway.

A valid default gateway address is 201.110.213.1. The network address (201.110.213.0) identifies the network as a whole, while the broadcast address (201.110.213.255) targets all hosts on the subnet simultaneously. Neither can be assigned as an interface address on a router gateway.

4). With the incorrect gateway, what communication may still work and what communication is likely to fail?

Local traffic within the same subnet (devices on 201.110.213.0/24) will still work via ARP and switch forwarding.

Any traffic destined for external networks or the Internet (such as queries to the DNS server at 8.8.8.8) will fail because packets cannot reach the gateway for routing.

5). A lab contains 25 student computers. Recommend static or dynamic addressing for the computers and name the service that would provide the addresses.

Using dynamic addressing to simplify IP management and avoid manual configuration errors across multiple endpoints.

Service: DHCP (Dynamic Host Configuration Protocol).

